



Study of formation and development of disturbance waves in annular gas–liquid flow



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ABSTRACT

Wavy structure of downward annular gas–liquid flow with liquid entrainment in 15 mm pipe was studied using high-speed laser-induced fluorescence technique. Measurements were performed near the inlet, which was organized as a tangential slot. Spatiotemporal records of film thickness were obtained over the first 100 mm of the pipe length in order to investigate formation and initial stages of development of the disturbance waves. It was shown that for high enough gas and liquid flow rates disturbance waves appear and start to dominate in the wavy structure of liquid film within the area of interrogation. Disturbance waves were found to be formed due to coalescence of small high-frequency waves appearing at the inlet. Similar mechanisms of formation of large waves with fast ripples on them were observed far downstream for waves near transition to entrainment and for ephemeral waves in flow regimes with entrainment. Significant individual acceleration of disturbance waves at the initial stage of their development was observed. Spectral analysis has shown strong energy transfer from high to low frequencies, which is in agreement to the proposed mechanism of waves formation.

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Introduction

In annular gas–liquid flow liquid flows as a film along the pipe wall in presence of high-velocity gas stream in the pipe core. This flow regime exists in a wide range of industrial equipment. Integral properties of the flow (such as pressure drop and heat transfer) are substantially different from those in single-phase flows and are difficult to predict because of complex interaction between the phases. The complexity increases at large enough gas and liquid flow rates when liquid droplets are torn from film surface and entrained into the gas core. Transition to entrainment drastically increases pressure drop and heat transfer. Complexity of the film flow increases due to appearance of multi-scale surface waves, deposition of droplets from the gas core and entrapment of gas bubbles into the film. The flow properties depend on flow rates and physical properties of gas and liquid, size, shape and orientation of the duct, configuration of the inlet and distance below the inlet.

The wavy structure of film surface in presence of liquid entrainment is dominated by so-called disturbance waves. These structures

represent large lumps of liquid travelling with large speed and carrying major fraction of liquid. Their height is several times larger than the thickness of base film layer between them. Surface of both disturbance waves and base film is covered with small-scale ripple waves. Presence of the disturbance waves is necessary for entrainment, since the droplets are torn from the tops of the disturbance waves. Modelling of annular flow requires understanding of structure of the disturbance waves and processes of their formation, evolution and interaction. The disturbance waves were intensively studied for over fifty years, but, due to numerous methodological difficulties, understanding of their nature is still far from perfect.

Two different interpretations of the disturbance waves are available in literature. According to the first interpretation, the disturbance waves are considered to be high-amplitude nonlinear waves with steep front, shallow rear slope and well-pronounced crest. This shape resembles the shape of solitary waves observable on thin falling films. The disturbance waves indeed have such appearance in the temporal records of film thickness obtained with pointwise low-resolution measurement techniques such as conductance probes (Chu and Dukler, 1974; Han et al., 2006; Al-Sarkhi et al., 2012; Zhao et al., 2013).

Application of spatially resolved methods results in different appearance and, hence, different interpretation of the disturbance waves. As it was first reported by Hewitt et al. (1990), the

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disturbance waves resemble relatively shallow plateaus covered by high-frequency large-amplitude ripple waves. By now this interpretation is supported by numerous observations using planar laser-induced fluorescence approach (Schubring et al., 2010; Farias et al., 2012; Zadrzil et al., 2014) and backlit side-view visualization (Pham et al., 2014; Setyawan et al., 2014; Pan et al., 2015). Alekseenko et al. (2008) studied both spatially and temporally resolved film thickness data and observed that the ripples are generated at the rear slopes of the disturbance waves, and may either accelerate and travel over the disturbance waves or decelerate and slide back to the base film. The scenario of evolution of a particular ripple was shown to be defined by the relative coordinate of the point of its inception. Such spatiotemporal behaviour could be explained by existence of eddy motion under the humps of the disturbance waves (Alekseenko et al., 2009). Recently, Hall Taylor et al. (2014) proposed a model for the disturbance waves in which eddy motion plays the central role. Zadrzil and Markides (2014) observed numerous eddies inside the large waves on thick falling films.

Choosing one of the two interpretations of the disturbance waves exerts important influence on the further line of thought. For example, different hypotheses about the mechanism of liquid entrainment correspond to different interpretations of the disturbance waves. Another – more relevant to the goal of the present paper – consequence is selection of a criterion of identification of the disturbance waves. This implies in the development of a suitable method to distinguish the disturbance waves among the waves on film surface and to understand whether the disturbance waves are present or not in particular flow conditions. The first interpretation implies that the differences between the disturbance waves and any other waves are mainly quantitative. This usually leads to development of semi-empirical amplitude-based criteria. E.g., a wave could be counted as a disturbance wave if its profile crosses the level of mean film thickness twice (Chu and Dukler, 1974) or if its height overcomes the mean film thickness more than 1.6 times (Zhao et al., 2013). Usage of the second interpretation leads to criteria related to presence of ripples on top of the disturbance waves. Based on the aforementioned observations of Alekseenko et al. (2008) it can be proposed to define disturbance wave as a wave generating ripples faster than the wave itself. This criterion will be used in further analysis throughout the paper.

Border of the area of existence of disturbance waves in flow maps is of hyperbola-like shape (Woodmansee and Hanratty, 1969; Andreussi et al., 1985). Area of existence of the disturbance waves encompasses the area of presence of entrainment; the former is only slightly larger than the latter. Exact position of the border depends on numerous factors. Flow orientation stronger affects the region of low gas and large liquid flow rates. The border is shifted depending on whether gravity exerts stabilizing or destabilizing influence on the waves. Thus, in vertical flow transition to the regimes with the disturbance waves occurs at lower gas velocities than in horizontal flows. In downward flow the disturbance waves are considered to appear even at zero gas velocities at large liquid flow rates (Webb and Hewitt 1975). In particular, Zadrzil and Markides (2014) reported existence of fast ripples on top of large waves at thick falling films. At moderate liquid flow rates and low gas velocities the disturbance waves coexist with waves typical to falling film regimes (Webb and Hewitt, 1975). Despite various quantitative differences, disturbance waves in flows of different orientation most likely belong to the same type of waves, since the processes of generation of fast and slow ripples are qualitatively the same in different conditions (Cherdantsev et al., 2014).

Liquid viscosity is more important in the region of low liquid and high gas flow rates; its increase shifts this part of the border towards even lower liquid flow rates. In this region, transition line is so steep that the idea of critical liquid Reynolds number Re_{Lcr} is often used for practical purposes (Ishii and Grolmes, 1975). Below Re_{Lcr} the film is covered by waves of two types, termed as “primary” and “secondary” waves (Alekseenko et al., 2009). Faster primary waves

generate slower secondary waves on their back slopes exactly as the disturbance waves generate the slow ripples. Nonetheless, the primary waves cannot be counted as the disturbance waves, since the secondary waves normally do not travel faster than the primary waves and, hence, do not climb over them.

In the disturbance waves region, two other types of waves – except disturbance waves and ripples – are sometimes specified (Sekoguchi and Takeishi, 1989; Sekoguchi and Mori, 1997). The huge waves are distinguished from the disturbance waves by even larger amplitude, size and velocity. They are mostly observed at low gas and large liquid flow rates and are supposedly related to transition from slug/churn to annular flow. The ephemeral waves appear at the rear slopes of the disturbance waves and lag behind them, moving over the base film. This behaviour is similar to that of slow ripples, but the ephemeral waves are much larger in amplitude than the slow ripples and are much rarer. Wolf et al. (1996) supposed that appearance of the ephemeral waves is related to excess film flow rate, which is gathered due to deposition of droplets from the gas core.

The flow is considered to be stabilized at the distances of several meters below the inlet (100–150 pipe diameters, according to Wolf et al., 2001). Between the inlet and the stabilization region local properties of the flow (including film properties) undergo essential changes. In particular, velocity of the disturbance waves grows with downstream distance (Azzopardi, 1997; Wolf et al., 2001), whereas the passing frequency essentially decreases (Hall Taylor and Nedderman, 1968; Zhao et al., 2013). The latter occurs due to coalescence of the disturbance waves, when a faster wave overtakes a slower one and a new wave is formed. As it was observed by Hall Taylor et al. (1963) and recently confirmed by Alekseenko et al. (2014), at large enough distances below the inlet the waves appearing due to coalescence most likely travel with the velocity of the faster coalescing wave. Without coalescence a disturbance wave may travel over large distances with constant velocity (Hall Taylor et al., 1963).

Formation of the disturbance waves occurs not far from the inlet (Hanratty and Engen, 1957; Hall Taylor et al., 1963; Zhao et al., 2013). Upstream the point where the disturbance waves are first observed, the film is covered by wavelets of high frequency and small amplitude (Zhao et al., 2013). It is still unclear if there exists any kind of interrelation between the initial waves and the disturbance waves appearing downstream. Hanratty and Hershman (1961) and Andreussi et al. (1985) developed a model in which formation of the disturbance waves occurs due to instability of gas-sheared films to long-wave perturbations. The calculated neutral stability conditions showed good agreement to the regime maps of existence of the disturbance waves in horizontal flow.

By present time, no direct experimental studies of process of formation of the disturbance waves were made. The goal of the present paper is to study the formation and the initial stages of development of the disturbance waves. For this, spatiotemporal measurements of film thickness were performed in the vicinity of the inlet.

Experimental setup and measurements technique

In the present work, downward adiabatic air–water flow is studied. The full scheme of the flow loop is described, e.g., in Alekseenko et al. (2012). Test section is a vertical cylindrical acrylic resin pipe with length of 1 m and inner diameter $d = 15$ mm (Fig. 1). Gas enters the working section through coaxial metallic tube with inner diameter of 13.4 mm and wall thickness 0.3 mm. Liquid enters through a ring-shaped tangential slot between the inner surface of the main pipe and the outer surface of the gas-feeding tube. Under such configuration of the inlet, liquid is introduced as a film, and annular flow regime takes place from the very beginning.

Superficial gas velocity, V_g , and liquid Reynolds number, Re_L , were chosen as quantities characterizing the flow rates of the phases.

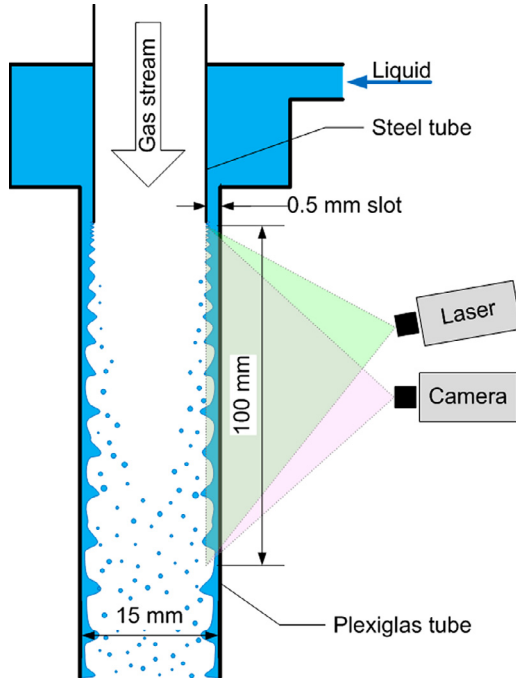


Fig. 1. Scheme of the inlet section.

The quantities were defined as

$$V_g = \frac{4Q_g}{\pi d^2}, \quad Re_L = \frac{Q_L}{\pi d \nu}. \quad (1)$$

Here Q_g and Q_L are volumetric gas and liquid flow rates and ν is kinematical viscosity of liquid. In our opinion, these two parameters provide the best generalization of annular flow data in pipes of different diameters and with different liquid viscosities (see, e.g., Figs. 5(b) and 6(b) in Alekseenko et al., 2013 and Fig. 8 in Alekseenko et al., 2014). The main set of experimental data of the present work belonged to the range of V_g between 15 and 57 m/s and to the range of Re_L between 140 and 400. An additional set of data at $Re_L = 30$ was obtained to create a reference matrix for the film thickness measurements. Also, for comparison purposes, the database of our earlier experiments conducted farther from the inlet was used. This database covers similar range of the flow rates, but also includes experiments with different liquid viscosity ($\nu = 1.9 \times 10^{-6} \text{ m}^2/\text{s}$) and pipe diameter ($d = 11.7 \text{ mm}$).

Film thickness measurements were performed with the method of brightness-based laser-induced fluorescence, which was used to study the wavy structure of gas-sheared liquid films in a number of our earlier papers (Alekseenko et al., 2008, 2014; Cherdantsev et al., 2014). A brief description will be given here.

Rhodamine 6G was used as a fluorescent matter. It was dissolved in working liquid in small concentration of 15 mg/l. In order to excite the fluorescence, continuous 2 W laser with wavelength 527 nm was used. The laser beam was converted to a vertical sheet with width of 1 mm, illuminating a longitudinal section of the pipe with length of 100 mm. This area of interrogation (AoI) started about 3 mm below the inlet to avoid bright reflections of laser light on metallic end of the gas-feeding tube. High-speed CCD camera "VideoSprint" produced by "VideoScan" was used to obtain instantaneous distributions of local brightness of the fluorescent light. Camera was equipped with orange low-pass filter with cut-off wavelength 550 nm. AoI was 'seen' by a line of 500 pixels of the camera matrix; so, each camera pixel is measuring brightness over a square with side of 0.2 mm. The camera worked at frame rate of 10 kHz; the exposure was varied within the range of 50–80 μs . At each time moment an instantaneous spatial distribution of local brightness of fluorescence over the longitudinal

coordinate was obtained. Temporal sequences of such distributions formed a matrix of local brightness, $J(x, t)$. For each regime point a record with duration of 2 s was obtained. Thus, each record represents a matrix of 500 by 20,000 values of the local brightness. Such matrices were transformed into matrices of local film thickness, $h(x, t)$, using the relation

$$J(x, t) = C(x)(1 - e^{-h(x, t)}) + D(x). \quad (2)$$

Here α is the coefficient of absorption of the laser light; k – the coefficient of reflection of the laser light from the liquid–gas interface (equal 2%); $D(x)$ is the dark level of the camera and $C(x)$ is the compensation matrix. The latter was constructed to compensate non-uniformity of the exciting laser light along the longitudinal coordinate. To create the compensation matrix, flow of thin liquid film falling under action of gravity was organized, and the time-averaged level of brightness registered by each pixel was considered to correspond to the Nusselt film thickness, h_N . The Nusselt formula is widely acknowledged as a reliable estimation of film thickness for moderate (up to 50) liquid Reynolds numbers. Its reliability was also confirmed in earlier experiments of our group, in which liquid layer of known thickness between the pipe wall and a coaxially mounted cylindrical insert (Alekseenko et al., 2012) was used to create the compensation matrix. $C(x)$ was defined as

$$C(x) = \frac{\langle J_N(x) \rangle - D(x)}{(1 - e^{-\alpha h_N})(1 + ke^{-\alpha h_N})}. \quad (3)$$

Here $\langle J_N(x) \rangle$ is the time-averaged brightness of the fluorescent light during recording the 'Nusselt' falling film at distance x . When all the components of Eq. (2) are known, $h(x, t)$ can be expressed as a function of $J(x, t)$. More detailed description of the recalculation and calibration procedures is given in Alekseenko et al. (2012).

Important difference between the described procedure and the present case is that the average thickness of the falling film is not constant in the vicinity of the inlet. The film thickness decreases gradually from 0.5 mm at the slot outlet to h_N . According to formula (5.2) in Alekseenko et al. (1994), at small liquid flow rates ($Re_L = 30$) the thickness is stabilized quite quickly and the local average film thickness will not exceed $1.01h_N$ at the distances of 4 mm below the inlet. Though this calculation is approximate, we assume that in the lower half of the AoI (i.e., distances larger than 53 mm below the inlet, which is 13 times larger than the calculated value) the thickness of the falling film can be considered stabilized. We also expect that the flow stabilization would be reached much faster under the action of strong gas shear for thin liquid films.

Thus, the two-step procedure for obtaining the reference signal was applied. At the first step the lower half (the distances 53–103 mm below the inlet) of the falling film flow record ($Re_L = 30$ and $V_g = 0 \text{ m/s}$) was used as a reference signal to measure the average film thickness at the same Re_L and large V_g . At the second step the record with $Re_L = 30$ and $V_g = 43 \text{ m/s}$ was used as a reference signal to recalculate all the other brightness records into the film thickness.

According to earlier estimations (Alekseenko et al., 2014), the overall error of the film thickness measurements can be estimated as 5–6% when studying the base film and the rear slopes of the disturbance waves. Under high slopes of the interface, which typically take place at the fronts of the fast ripples on top of the disturbance waves, total internal reflection of the exciting light is likely to occur. In such a case the local intensity of exciting light increases inside the film, which causes increase in the intensity of the fluorescent light and, hence, in measured local film thickness. An example of the film thickness record with such distortions is given in Fig. 4 in Alekseenko et al. (2014). This is the strongest shortcoming of the brightness-based LIF technique. Near the inlet, high slopes can be expected not only at the fast ripples' fronts, but also at the slopes of high-frequency waves appearing near the inlet (see the top plot in Fig. 9 in Zhao et al., 2013). Thus, the height of such waves may be

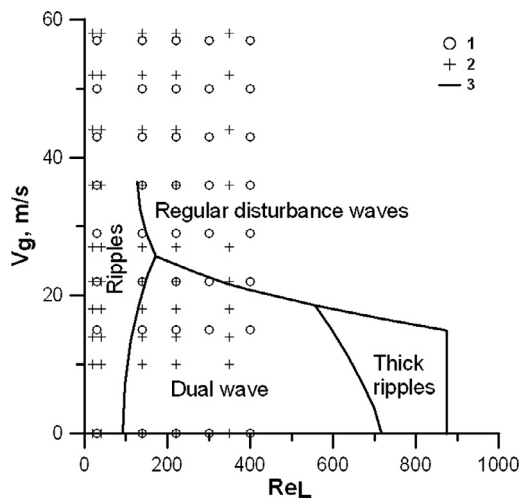


Fig. 2. Set of flow conditions used in the present experiments (1) and in the earlier air–water experiments with $d = 15$ mm farther from the inlet (2). Borders of the wavy patterns according to Webb and Hewitt (1975) are also shown (3).

overestimated by the technique. Regarding this, the main emphasis in the present paper is made on qualitative description of transition from the initial waves to the disturbance waves and on measurements of the velocity and the frequency of the waves, instead of the amplitude characteristics.

Formation of disturbance waves

In the present section, qualitative observations of spatiotemporal evolution of the film surface will be presented. First, the wavy patterns for different flow regimes far from the inlet will be described. Second, the formation of disturbance waves near the inlet will be shown. Finally, similar phenomena of generation of waves far from the inlet will be demonstrated.

Flow regimes and waves behaviour far from the inlet

The set of flow regimes studied in present work is shown by circles in Fig. 2. Before starting to analyse the behaviour of waves in the vicinity of the inlet, it is reasonable to understand the spatiotemporal behaviour of waves farther downstream, where the inlet effects are expected to be weak. For this purpose, we used the data obtained earlier (see, e.g., Alekseenko et al., 2013, 2014) in the same pipe by the same technique, but at the downstream distances $x = 500$ – 620 mm. These flow regimes are shown by crosses in Fig. 2.

The solid lines in Fig. 2 represent the borders of the wavy patterns obtained by Webb and Hewitt (1975), transformed into the variables used in the present work. In that paper, air–water downward flow was studied in a pipe with inner diameter $d = 38.2$ mm at the downstream distances from 2 to 15 m below the inlet. According to this map, in our main data set the regimes of “regular disturbance waves” and “dual waves” should be expected. The latter is characterized by co-existence of the disturbance waves typical to the large gas velocity regimes and the waves typical to falling film regimes.

With the present configuration of LIF technique it is possible to obtain the spatiotemporal records of film thickness $h(x, t)$. Each string of such matrix is an instantaneous profile of film thickness along longitudinal coordinate; each column of the matrix is a temporal record of film thickness at fixed downstream distance. Using such a matrix, continuous spatiotemporal trajectory of each wave can be detected. We will present such matrices in form of the images (see Figs. 3–9, 12). Each image consists of pixels, corresponding to the elements of the matrix $h(x, t)$, with brightness of each pixel directly proportional

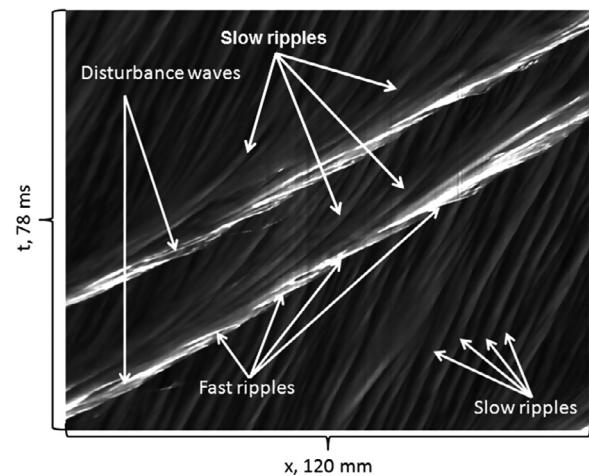


Fig. 3. Fragment of film thickness matrix for $x = 500$ – 620 mm. $Re_L = 140$, $V_g = 27$ m/s.

to the value of local film thickness in this element of the matrix. In the majority of such figures, white corresponds to the local film thickness of 1 mm and larger. In some figures the coefficients of proportionality are different: white corresponds to $h \geq 0.5$ mm in Fig. 8 and to $h \geq 0.25$ mm in Figs. 5 and 9. The vertical coordinate corresponds to time; the horizontal coordinate – to the longitudinal distance. The flow direction is from left to right. If not specified, the data are shown for the air–water flow in 15 mm i.d. pipe.

Fig. 3 shows example fragment of matrix $h(x, t)$ obtained at relatively large ($x = 500$ – 620 mm) distance below the inlet. At such distances the disturbance waves are formed already. Two of them are visible in Fig. 3. Each disturbance wave crosses the area of interrogation with constant velocity (which is proportional to the slope of the spatiotemporal trajectory of a wave to the t -axis). Velocities of individual disturbance waves may be different. The disturbance waves are separated by the areas of thin residual layer of liquid, known as “base film”. The base film is covered with the slow ripple waves. As it was first observed by Alekseenko et al. (2008), all the slow ripples appear at the rear slopes of the disturbance waves. The rear slopes in x – t representation correspond to the upper (or left) borders of the waves. Just after inception, the slow ripples travel with the velocity close to that of the disturbance waves; but then they gradually decelerate and slide to the base film behind the “parent” disturbance wave. To check this, the reader is encouraged to track the marked slow ripples back in time (downwards) to their inception points. Finally, the slow ripples travel over the base film with constant low speed until the following disturbance wave absorbs them. Surface of the disturbance waves is covered by the fast ripples which are also generated at the rear slopes of the disturbance waves (see Alekseenko et al., 2008). The fast ripples travel from the rear slope towards the front of the disturbance wave and eventually disappear. They disappear either because they are disrupted by the gas shear into droplets or due to the decay at thin base film in front of the disturbance waves (Cherdantsev et al., 2014).

Fig. 4 shows comparison of long records of the film thickness obtained near the transition between the wavy regimes at $Re_L = 350$. At $V_g = 18$ m/s (Fig. 4a) the disturbance waves separated by thin base film are clearly visible. At $V_g = 14$ m/s (Fig. 4b) the wavy structure is highly irregular; the areas of thin base film covered with the slow ripples are intermittent with the areas of thick film with irregular long-wavelength perturbations of thickness. This is very similar to the description of the ‘dual-wave’ regime observed by Webb and Hewitt (1975). In our pipe the transition between the ‘dual-wave’ and the ‘regular disturbance wave’ regimes occurs at slightly lower gas velocities: regular disturbance waves are first observed at $V_g = 22$ m/s

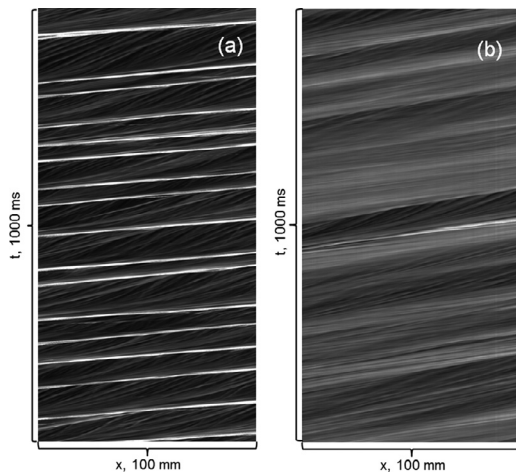


Fig. 4. Spatiotemporal records of film thickness for $Re_L = 350$ and $V_g = 18$ m/s (a) and $V_g = 14$ m/s (b). $x = 500$ – 620 mm.

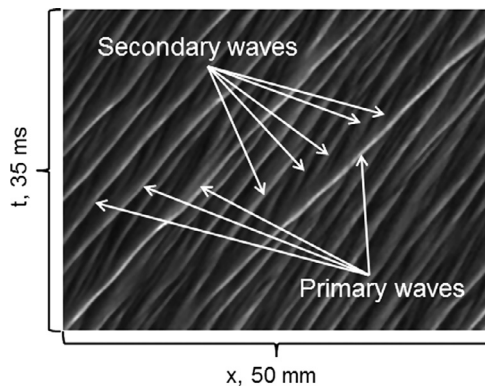


Fig. 5. Example of spatiotemporal behaviour of primary and secondary waves in absence of liquid entrainment. $Re_L = 20$, $V_g = 44$ m/s, $x = 500$ – 620 mm.

for $Re_L = 140$ and at $V_g = 18$ m/s for $Re_L = 220$ and 350 . The discrepancy is most likely linked with the difference in the pipe diameters. We should also note that the regime point $V_g = 27$ m/s and $Re_L = 140$, shown in Fig. 3, definitely belongs to the ‘regular disturbance waves’ region and not to the ‘ripples’ region as it would follow from the map shown in Fig. 2.

Finally, at small Re_L and large V_g we can observe wavy pattern which was called ‘ripple regime’ by Webb and Hewitt (1975). Example of a spatiotemporal record for this regime is shown in Fig. 5. As it was first time shown by Alekseenko et al. (2009), in such regimes it is reasonable to ‘separate’ the observed ‘ripples’ into two types: fast long-living ‘primary waves’ and slower short-living ‘secondary waves’ generated at the rear slopes of the primary waves. Interrelations between the primary and the secondary waves remind the interrelations between the disturbance waves and the slow ripples (see Fig. 3). Besides quantitative differences in velocity, size and height of the disturbance waves and the primary waves, there is a qualitative difference consisting in that the primary waves do not generate “fast secondary waves”.

The secondary waves in flow regimes without entrainment are very similar to the slow ripples in flow regimes with entrainment. They are both generated at the rear slopes of larger and faster waves and travel with low speed over the base film. The differences between them are mostly quantitative and are mainly defined by the liquid flow rates, which is much larger in the case of slow ripples. The main qualitative difference is that the velocity of the secondary waves is nearly constant starting from the point of inception, whereas the slow ripples start with velocity close to that of parent disturbance waves

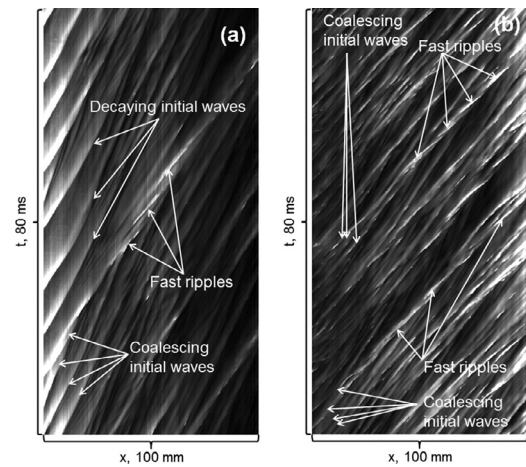


Fig. 6. Fragment of film thickness matrix near the inlet. $Re_L = 220$, $V_g = 22$ m/s (a); $Re_L = 400$, $V_g = 29$ m/s (b).

and then decelerate. This might be explained by larger velocity and thickness of the film at the inception point in case of entrainment regimes

To summarize this section, the following statements should be made:

- (1) The disturbance waves exist in the whole investigated range of flow rates ($V_g = 15$ – 57 m/s and $Re_L = 140$ – 400).
- (2) At low V_g they coexist with the waves typical to the falling film regions (the ‘dual-wave’ regime).
- (3) Presence of the fast ripples should be used as a criterion of a disturbance wave.

Formation of the disturbance waves near the inlet

With the results obtained in subsection “Flow regimes and waves behaviour far from the inlet”, it is possible to analyse the data obtained in the vicinity of the inlet. Fig. 6a shows an example fragment of $h(x, t)$ matrix, obtained in the present experiments at the distances $x = 3$ – 103 mm below the inlet at relatively low gas and liquid flow rates. The very vicinity of the inlet is covered with the initial waves of relatively high frequency. Their amplitude is very high according to LIF data, but it might be significantly overestimated due to the optical distortions caused by total internal reflection of the exciting light at the steep front slopes (see, e.g., Alekseenko et al., 2014). These initial waves are rather irregular by the individual velocity and by the separation time. This irregularity may lead to coalescence of the neighbouring waves, forming new waves with different properties and spatiotemporal behaviour.

The waves, appearing as a result of the coalescence of the initial waves, are covered with the fast ripples. Thus, according to the proposed criterion, even at the moderate gas and liquid flow rates the disturbance waves appear within the area of measurements. It should be also noted that in absence of coalescence the initial waves tend to decelerate, losing amplitude and changing in shape. This is possibly related to rapid thinning of the film starting from the slot thickness value due to the action of the gas shear. With increasing liquid and, especially, gas flow rates the initial waves become much faster and shorter. Since the separation time is greatly reduced and the absolute scatter in velocity of individual waves is increased, the average distance required for the coalescence of neighbouring initial waves decreases. As a result, the disturbance waves are formed closer to the inlet and in larger quantities, but the mechanism of their formation remains the same (Fig. 6b).

Example fragments of $h(x, t)$ matrices for large gas velocities are shown in Fig. 7. In contrast to Fig. 6a, the disturbance waves certainly

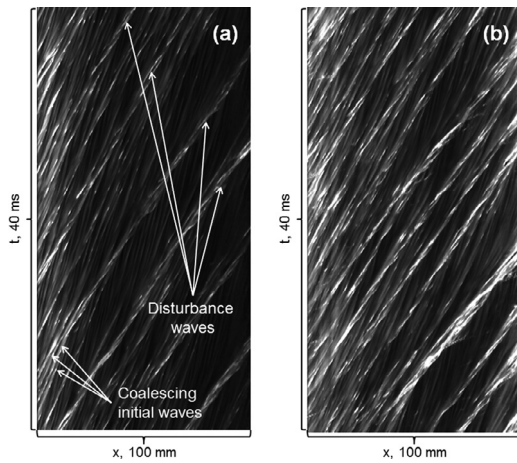


Fig. 7. Fragment of film thickness matrix near the inlet. $V_g = 50$ m/s. $Re_L = 140$ (a); $Re_L = 300$ (b).

dominate in the wavy structure starting from a few centimetres below the inlet. Only the disturbance waves separated by thin base film are observed in the farther half of the area of interrogation. Of course, the coordinates of formation of the individual disturbance waves are different, since the process of interaction of the initial waves is probabilistic. As it could be seen from comparison of Fig. 7a and b, the frequency of the disturbance waves noticeably increases with the liquid flow rate.

To the best of our knowledge, the observed mechanism of formation of the disturbance waves via coalescence of the initial high-frequency perturbations is described for the first time. Zhao et al. (2013) observed that only ‘ripples’ existed on film surface prior to appearance of the disturbance waves, but these ripples were not linked to appearance of the disturbance waves downstream.

The observation on the crucial role of the initial small-scale waves seems contradictory to the model of long-wave instability employed by Hanratty and Hershman (1961) and Andreussi et al. (1985). This model showed good overall agreement with the experimentally observed conditions of the existence of the disturbance waves. Moreover, Hanratty and Hershman mentioned that addition of a surfactant, which suppressed the short waves, did not eliminate the appearance of the disturbance waves. We suppose that there is no real contradiction between the two approaches. Instability of film surface to large-scale perturbations is necessary for existence of the disturbance waves. But the coalescence of several short waves is an efficient way of creating a large-scale perturbation of the film surface. Such perturbations grow due to long-wave instability and reach large amplitude values much faster than the initial long-wave perturbations of a smooth film surface would do.

The observed spatiotemporal behaviour also resembles the wavy pattern predicted by Chang et al. (2002) who modelled evolution of a system of nonlinear solitary waves on falling films in absence of gas flow. In their model, due to broad-band excitation noise, fraction of the waves developed slightly larger amplitudes and, hence, larger velocities. This led to coalescence of the neighbouring solitary waves, causing decrease in the waves frequency and appearance of the waves of essentially larger amplitudes and velocities. In our case this effect is enhanced by qualitative change in properties of the waves appearing due to coalescence. Explanation of this qualitative change of the wave is not entirely clear, but it is certainly accompanied by gaining ability to generate the fast-moving waves on the wave’s rear slope. Bearing in mind that this ability is possibly related to the existence of an eddy (Alekseenko et al., 2009) or several eddies (Zadrazil and Markides, 2014) under the hump of the wave, it can be hypothesised

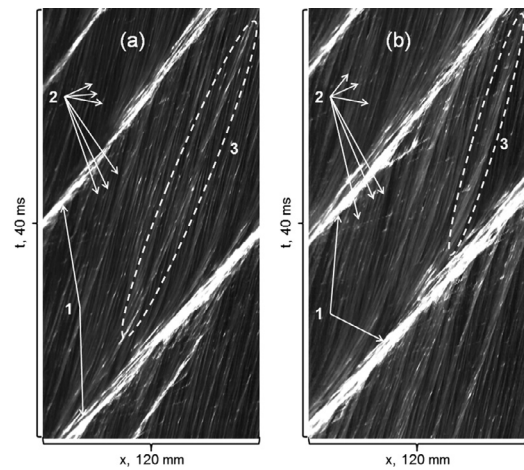


Fig. 8. Formation of ephemeral waves at the rear slopes of disturbance waves. (1) Disturbance waves; (2) slow ripples; (3) ephemeral wave. $Re_L = 142$, $V_g = 58$ m/s (a); $Re_L = 220$, $V_g = 52$ m/s (b). $x = 500$ – 620 mm.

that the coalescence initiates or enhances the eddy motion in the resulting wave.

Formation of waves with fast ripples far from the inlet

One could suggest that the mechanism of formation of the disturbance waves observed in section “Formation of the disturbance waves near the inlet” is valid only in the vicinity of the inlet and may be realized only with the inlets of certain configuration and dimensions. In order to investigate how universal the described mechanism is, search for the waves with the fast ripples, appearing far from the inlet, was performed over a wide range of flow conditions.

The first such case is the formation of so-called “ephemeral” waves in the flow regimes with liquid entrainment far from the inlet. As mentioned in “Introduction” section, such waves appear at the rear slopes of the disturbance waves and are different from the slow ripples on a number of quantitative characteristics. We can observe the ephemeral waves in our data; their frequency of occurrence grows with gas and liquid flow rates. The examples of formation and spatiotemporal behaviour of the ephemeral waves are shown in Fig. 8. In each part of this figure, an ephemeral wave is marked by a dashed ellipse with number 3. It can be seen that the ephemeral waves are covered with the fast ripples and most likely belong to the same type of waves as the disturbance waves do. The ephemeral waves are formed due to coalescence of the slow ripples, appearing at the rear slopes of the ‘parent’ disturbance waves. The ephemeral waves might play an important role at large downstream distances, where frequency of the disturbance waves decreases. Supposedly, an ephemeral wave can evolve into a ‘proper’ disturbance wave, if it has enough space to develop and accelerate before the following disturbance wave could absorb it.

The second case was observed at low liquid and large gas flow rates, where no liquid entrainment was detected with a sampling probe during long-time runs. As mentioned above (see Fig. 5), in such flow conditions the primary waves normally generate only the slow secondary waves on their rear slopes. Nonetheless, at very large V_g when liquid flow rates approach the critical Reynolds number, the waves covered with the fast ripples are occasionally observed. Such waves are much rarer than the ‘regular’ primary waves. The frequency of their appearance grows with increasing gas velocity, liquid flow rate and liquid viscosity. These waves are also more frequent in pipes of smaller diameter. Examples of such waves are shown in Fig. 9. It can be seen that such waves are formed due to coalescence of the primary waves.

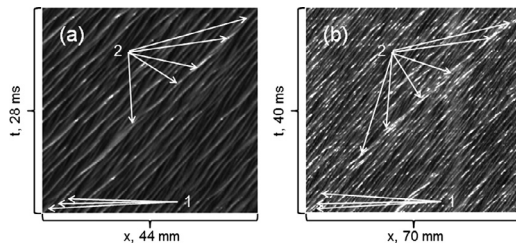


Fig. 9. Formation of waves covered with fast ripples in annular flow without entrainment. (a) $Re_L = 40$, $V_g = 58$ m/s, water, $d = 15$ mm, $x = 500$ – 620 mm. (b) $Re_L = 32$, $V_g = 70$ m/s, water–glycerol solution with $\nu = 1.9 \times 10^{-6}$ m²/s, $d = 11.7$ mm, $x = 400$ – 500 mm. (1) Coalescing primary waves; (2) fast ripples.

Despite the presence of the waves with fast ripples, no entrainment was observed in these flow regimes. We can conclude that the presence of the fast ripples is not sufficient for entrainment to occur. Obviously, there is a range of flow conditions in which the fast ripples can be generated, but cannot yet be broken into droplets. Besides, in these flow conditions the waves with fast ripples do not live long, in contrast to the case of larger Re_L . This observation is in agreement with the above considerations on the role of film stability to large-scale perturbations (section “Formation of the disturbance waves near the inlet”).

To conclude, all the waves generating fast ripples are formed due to coalescence of shorter waves. In different cases different kinds of short waves serve as ‘material’ for the waves with fast ripples. Formation of the disturbance waves near the inlet requires coalescence of the initial waves, which are presumably the waves of maximum growth appearing due to Kelvin–Helmholtz instability. Formation of the ephemeral waves occurs due to coalescence of the slow ripples at the rear slopes of the disturbance waves. Formation of the waves with fast ripples near transition to entrainment occurs due to coalescence of the primary waves.

Coalescence of the short waves does not necessarily lead to formation of a wave with fast ripples, since coalescence may occur in nearly any flow regime. Some additional circumstances are required for a new wave to be able to generate the fast ripples and possibly have an eddy inside. Proper identification of such circumstances requires either an extensive systematic investigation of coalescence events, or theoretical analysis, or both.

Initial stages of development of disturbance waves

In this section, the initial stages of downstream development of the disturbance waves will be quantitatively investigated. As it was shown, for the majority of the investigated flow rate combinations the disturbance waves already dominate in the wavy pattern within the area of interrogation. The obtained spatiotemporal records of film thickness are equivalent to 500 temporal records of film thickness with duration of 2 s each recorded with frequency of 10 kHz, obtained by ‘probes’ placed 0.2 mm apart at the distances 3–103 mm below the inlet. Commonly used signal processing techniques such as cross-correlation analysis and spectral analysis could be applied to such records in order to study the downstream change of ‘local’ average properties of the waves. Besides, an algorithm of automatic identification of characteristic lines of the disturbance waves was applied to the present data in order to validate and further improve the results.

Velocity of developing disturbance waves

In order to measure the average velocity of the disturbance waves at certain downstream position, the cross-correlation technique was applied to the temporal records of film thickness. For every coordinate x_i from the inlet the cross-correlation function was calculated for

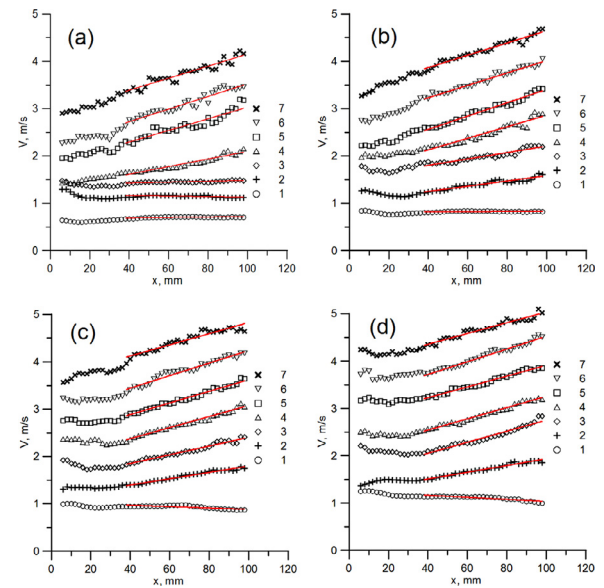


Fig. 10. Evolution of cross-correlation velocity of waves with the downstream distance. $Re_L = 140$ (a); 220 (b); 300 (c); 400 (d). Gas velocities change from 15 to 57 m/s with increment of 7 m/s (1–7).

two temporal records obtained in points $x_i + dx$ and $x_i - dx$, where dx is equal to 10 pixels or 2 mm. Velocity is equal $2dx$ divided by the time delay corresponding to the maximum of cross-correlation function. To increase accuracy of the time delay measurements, vicinity of the maximum was approximated by a cubic polynomial, and the position of the extremum point of the approximation was identified as the time delay. Evolution of the cross-correlation velocity of waves, V_{corr} , with the downstream distance for different flow regimes is shown in Fig. 10.

It can be seen that at high enough gas and liquid flow rates the cross-correlation velocity grows with the downstream distance. The growth starts closer to the inlet at larger gas velocities; slope of the dependence increases with the gas velocity. At the lowest gas velocities ($V_g = 15$ – 29 m/s for $Re_L = 140$ and $V_g = 15$ m/s for larger Re_L), no increase in V_{corr} is observed; on the contrary, the graph might even show slight deceleration. At moderate gas velocities deceleration is observed only in the very beginning and turns into acceleration at larger distances. This initial deceleration might be related to the above-mentioned decay of the initial waves without coalescence, which is further compensated by development of successful disturbance waves. This is quite clear from comparison with the spatiotemporal matrices of film thickness (e.g., Fig. 6a). The minimum in local velocity in the line (2) in Fig. 10b ($Re_L = 220$, $V_g = 22$ m/s) corresponds to the area where the initial waves decay in amplitude and decelerate (as marked in Fig. 6a). Later, large-scale disturbance waves are formed and maximum in the cross-correlation function corresponds to their velocity.

According to our observations, start of V_{corr} increasing corresponds to the beginning of the area where disturbance waves dominate in the overall wavy structure. At low V_g the disturbance waves are very rare and their contribution into the cross-correlation function is lower than that of slower initial waves.

To study the acceleration of the disturbance waves, $V_{corr}(x)$ was linearly approximated in the range $x = 40$ – 100 mm, where this dependence is approximately linear for all regime points. Slope of this approximation gives ‘spatial’ acceleration of the waves, which is shown in Fig. 11a. The spatial acceleration grows with gas velocity until it reaches limiting value of approximately 12 s^{-1} . It is interesting to relate the measured acceleration in space, a_x , to the acceleration of the disturbance waves in time, a_t , since the latter is expected to be

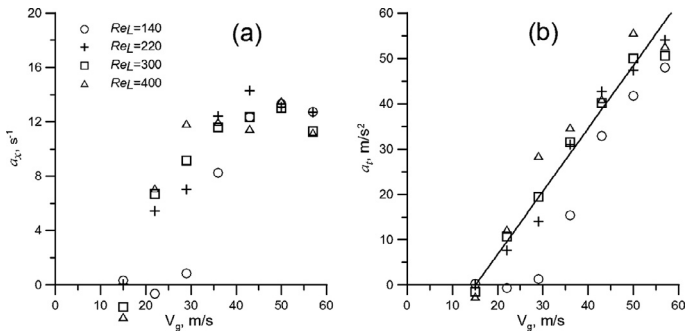


Fig. 11. Spatial (a) and temporal (b) acceleration of disturbance waves at $x = 40$ – 100 mm below the inlet.

tied with the forces (such as interfacial shear stress) acting on the disturbance waves. For a material point moving with constant temporal acceleration the velocity is expected to show square-root dependence on x :

$$V = \sqrt{V_0^2 + 2a_t(x - x_0)}. \quad (4)$$

In our case $x_0 = 40$ mm and $V_0 = V(x_0)$. The linear growth of $V_{corr}(x)$ observed experimentally seems contradictory to formula (4). One could assume that a_t also grows linearly with x . On the other hand, the used distance range is small, and it is likely that the deviation of the square-root dependence from linearity is still within the uncertainty of measurements. With the available data, it is difficult to check whether a_t is a constant, but its average value within the investigated distance range can be estimated by dividing the increase in velocity across the distance by the time required for a disturbance wave to pass this distance. With such approach, a_t can be estimated by multiplying a_x by the average velocity in this range. The value of the linear approximation of $V_{corr}(x)$ at $x = 70$ mm was taken as the average velocity, as if a_t was a constant. Results on acceleration of the disturbance waves in time for all regime points are plotted in Fig. 11b.

For the majority of regime points a_t linearly grows with V_g and nearly does not depend on Re_L . The lowest liquid flow rate ($Re_L = 140$) deviates from this dependence, but even in this case a_t approaches the common line at large V_g . This line is built using all the data points for $Re_L = 220$ – 400 and $V_g = 22$ – 57 m/s. It can be described as

$$a_t = 1.4(V_g - V_{g0}) \quad (5)$$

Here $V_{g0} = 15$ m/s; this value roughly corresponds to transition between the ‘dual-wave’ and ‘regular disturbance wave’ patterns for $Re_L = 220$ and higher (see Figs. 2 and 4).

The observed increase in the average velocity may occur either due to individual acceleration of each disturbance wave or due to coalescence of the disturbance waves. As mentioned above, in process of coalescence the resulting wave normally travels with the same velocity as the faster of the two coalescing waves (Alekseenko et al., 2014). In this case growth of the average velocity occurs due to elimination of the slowest waves from the total distribution. The distinction between these explanations gets more complicated since they do not exclude each other. Besides, the accelerations are measured over relatively short distances and times, so it is uneasy to detect them visually. Estimations show that the maximum difference between the parabolic trajectory with the measured acceleration and the linear approximation of this trajectory would not overcome 2 mm, which is several times smaller than the longitudinal size of a disturbance wave. To shed some light on the nature of acceleration, spatiotemporal trajectories of individual disturbance waves were identified in the region $x = 40$ – 100 mm using the algorithm developed by our group earlier (Alekseenko et al., 2014). The algorithm is described in details in that paper and it is exactly the same in the present case.

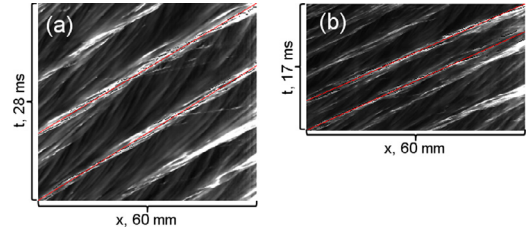


Fig. 12. Examples of automatically identified trajectories of two neighbouring disturbance waves (red lines). (a) $Re_L = 300$, $V_g = 43$ m/s; (b) $Re_L = 400$, $V_g = 57$ m/s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

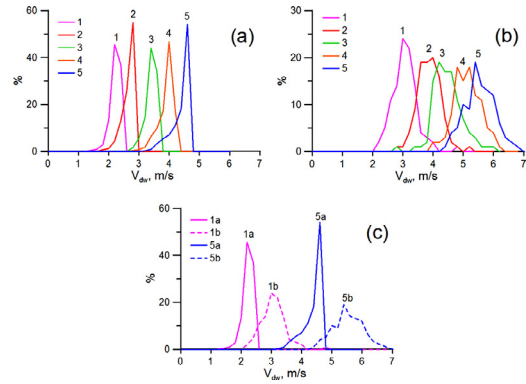


Fig. 13. Normalized velocity distributions of individual disturbance waves. (a) $x = 40$ – 100 mm, $Re_L = 400$; $V_g = 29$ m/s (1), 36 m/s (2), 43 m/s (3), 50 m/s (4), 57 m/s (5). (b) $x = 500$ – 620 mm, $Re_L = 350$, $V_g = 27$ m/s (1), 36 m/s (2), 44 m/s (3), 52 m/s (4), 58 m/s (5). (c) Comparison of velocity distributions at different distances.

Efficiency of the algorithm was visually checked using the auxiliary images with obtained characteristic lines of two neighbouring disturbance waves, superimposed on the fragment of matrix of film thickness (examples are shown in Fig. 12). It was found that at high gas and liquid flow rates, where the disturbance waves are dominant, the number of disturbance waves missed by the algorithm is always less than 10% of the total number of disturbance waves. Besides, the majority of the missed waves correspond to the slow disturbance waves which are absorbed within the Aol. No ‘false’ waves were produced by the algorithm. So, in this range of conditions the algorithm can be considered as an acceptable instrument to detect and study the individual disturbance waves. Obtained distributions of the disturbance waves by velocity for $Re_L = 400$ are shown in Fig. 13a. For comparison, the distributions obtained at $x = 500$ – 620 mm below the inlet at $Re_L = 350$ and similar values of V_g from Alekseenko et al. (2014) are given in Fig. 13 b. In both cases the bin width is 0.2 m/s and the number of the waves in each bin is normalized by the total number of the waves.

At $x = 500$ – 620 mm the velocity distributions are roughly symmetric around the average (Fig. 13b); the standard deviation of such distributions was found to be nearly constant (about 0.45 m/s) in a wide range of flow conditions. Velocity distributions, obtained near the inlet, are different (Fig. 13a). First, they are essentially asymmetric, with long tail protruding into the area of low velocities. Possibly, it is related to low velocity of the waves which have just appeared; such waves are expected to be promptly absorbed downstream. Second, the standard deviation is much smaller than that measured downstream, even if the ‘slow tail’ is taken into account. The most probable value of velocity corresponds well to the cross-correlation velocity of waves in the middle of the processed distance (at $x = 70$ mm). This means that the slow disturbance waves from the low-velocity tails of the distributions do not affect the velocities measured by cross-correlation technique.

For the velocities larger than the most probable value the distribution drops sharply as if it was cut artificially. This is not the case: there is indeed a limitation for maximum velocity of individual waves in the algorithm, but it is equal to $1.5 V_{corr}$; this value is much larger than the maximum observed value of the individual velocity. Actually, the shape of the upper half of the main peak indicates that the main part of each velocity distribution is very narrow and roughly symmetric.

The cross-correlation velocity of disturbance waves measured near the inlet is 20–40% smaller than that measured 500 mm downstream (compare the most probable velocities in Fig. 13c). Moreover, the velocity distributions obtained at the different downstream distances, nearly do not overlap. This implies that the coalescence mechanism itself cannot explain the observed growth of velocity.

It is interesting to estimate how fast the waves with such acceleration would reach the velocity values, experimentally measured at $x = 500$ – 620 mm. Both linear extrapolation of $V(x)$ dependence (as if a_x was constant) and calculation using formula (4), as if a_t was constant, show that it would happen at approximately 200 mm ($\sim 13d$) below the inlet for the majority of flow regimes. The difference between the two methods of calculation is indeed rather small (about 1–2 cm) in this range. The simplest idea is to divide the process of velocity growth into two stages. Strong initial increase in the average velocity occurs due to individual acceleration of the disturbance waves at the first 10–15 pipe diameters from the inlet. After this, the velocity is assumed to increase only due to the coalescence mechanism, having only slight spatial acceleration. For such development, a simple model was made by Alekseenko et al. (2014), which showed gradual increase of the average velocity and decrease of the standard deviation.

In order to make sure that the individual acceleration really takes place, an additional check using the automatic algorithm was performed. For each disturbance wave a set of spatiotemporal coordinates $t_m(x)$ was obtained. Here t_m is the time moment corresponding to the maximum film thickness value in the temporal vicinity of the characteristic line at the coordinate x . Examples of such sets of $t_m(x)$ are marked by the black dots in Fig. 12. The temporal acceleration of an individual disturbance wave was estimated as double value of the first coefficient of parabolic approximation of $x(t_m)$. This method yields large scatter of the measured acceleration values (typical standard deviation was smaller but comparable to the average value) since the length of a disturbance wave is several times larger than the deviation of a parabolic trajectory from its linear approximation. Nonetheless, this method seems to be the most reliable instrument to estimate the individual accelerations of the disturbance waves in the present experiments. It shows that the average of all individual accelerations within a flow regime is always of the same (positive) sign and of the same order of magnitude as the average acceleration measured using the cross-correlation analysis.

Frequency of developing disturbance waves

Spectral analysis was applied to the temporal records of film thickness to investigate the downstream evolution of frequency of maximum power in the power spectral density (PSD). For every coordinate x_i from the inlet, the temporal records obtained by 11 pixels (from $x_i - 1$ mm to $x_i + 1$ mm) were processed. Each record consisting of 20,000 points was divided into 8 equal parts. Thus, for each value of x_i , 88 records of film thickness with duration of 250 ms each were processed. Application of fast Fourier transform to each record gives spectral amplitudes of 1250 frequency values with step of 4 Hz. The obtained 88 spectra were squared to get the power spectra and averaged into a single one. This spectrum was smoothed with running average filter with window size of 7 points. Then the power spectrum was normalized by its total energy.

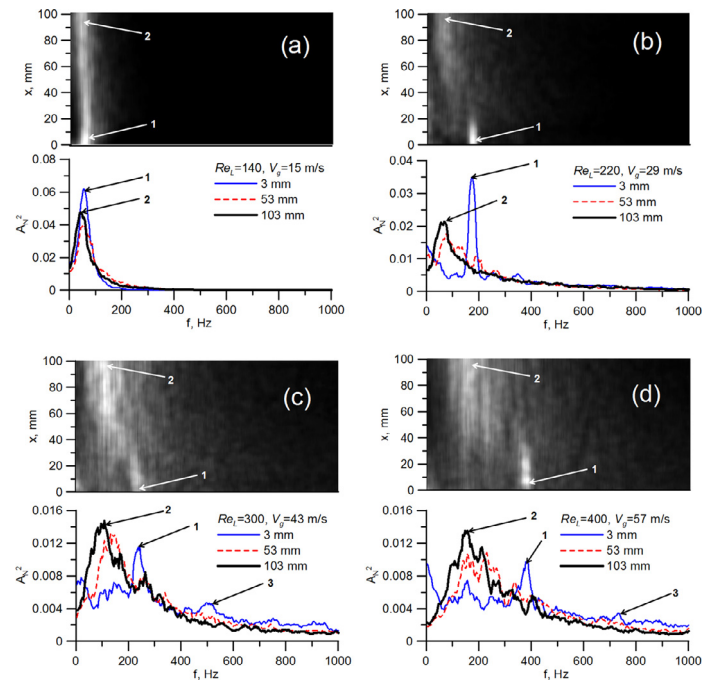


Fig. 14. Spatial evolution of normalized power spectra: (a) $Re_L = 140$, $V_g = 15$ m/s; (b) $Re_L = 220$, $V_g = 29$ m/s; (c) $Re_L = 300$, $V_g = 43$ m/s; (d) $Re_L = 400$, $V_g = 57$ m/s. The numbers mark: (1) frequency of maximum power at $x = 3$ mm; (2) the same at $x = 103$ mm; (3) monotonically growing mode at $x = 3$ mm (see Fig. 15). (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

Examples of spatial evolution of PSD are shown in Fig. 14 for four different combinations of liquid and gas flow rates. Each part of Fig. 14 consists of two parts: the upper part is a graphical representation of the PSD matrix. Each string of such a matrix is a PSD obtained at fixed downstream position. Local brightness is directly proportional to the value of spectral power. The lower parts show normalized PSD for three coordinates ($x = 3$ mm, 53 mm and 103 mm).

In the regimes where disturbance waves do not dominate, such as the one shown in Fig. 14a, PSD has one well-pronounced peak, marked by (1). We suppose that the frequency of this peak, f_{max} , corresponds to the frequency of the initial waves. In the frequency range $f > f_{max}$ the spectral power, $A_N^2(f)$, decreases in exponential manner. In the range $f < f_{max}$ $A_N^2(f)$ does not reach zero values; this is related to irregular character of waves, in particular, to large variation of the initial waves by amplitude and time separation. It can be seen from the upper part of Fig. 14a that the peak in PSD gradually shifts towards lower frequencies with the downstream distance, and the shape of PSD does not undergo essential changes. Position of the peak at the end of the AoI is marked by (2).

For the regimes with domination of the disturbance waves (Fig. 14b–d) the downstream evolution of PSD is different. Frequency of the initial peak (marked as 1 in these figures) is normally larger due to larger gas and liquid flow rates. In the frequency range between zero and the initial peak the PSD is stretched into a roughly uniform plateau. Spectral amplitude of the initial peak decays with the downstream distance and eventually this peak cannot be distinguished. The distance over which the peak is distinguishable decreases with gas velocity. This decrease is demonstrated in Fig. 15 for $Re_L = 140$. Parallel to the decay of the initial peak, secondary peak with lower frequency (marked as 2) develops in the central part of the plateau. This peak grows in normalized amplitude and gradually shifts towards the lower frequencies with the downstream distance. Thus, in these cases, there is no gradual drift from peak (1) into peak (2), as can be seen from the upper parts of Fig. 14b–d.

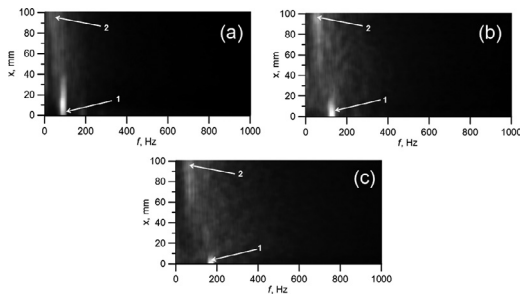


Fig. 15. Spatial evolution of normalized power spectra, $Re_L = 140$: (a) $V_g = 22$ m/s; (b) $V_g = 29$ m/s; (c) $V_g = 36$ m/s. The numbers mark: (1) frequency of maximum power at $x = 3$ mm; (2) the same at $x = 103$ mm.

Transition between the two scenarios of downstream evolution of PSD (with and without gradual change of the main peak) occurs between $V_g = 15$ m/s and $V_g = 22$ m/s. At $V_g = 15$ m/s there is gradual change for all values of Re_L ; for higher V_g , there is parallel decay of the initial peak and growth of the secondary peak for all Re_L . So, the latter scenario occurs in roughly the same regimes where the average velocity of waves grows with the downstream distance. Flow regimes with $Re_L = 140$ and $V_g = 22$ –29 m/s are exceptions here. Possibly, at low Re_L larger distances are required for domination of the disturbance waves.

We suppose that the initial peak (1) corresponds to the characteristic frequency of the initial waves, whereas the peak (2) describes the disturbance waves. In this case, the energy transfer from high to low frequencies corresponds to the described above process of coalescence of high-frequency initial waves into lower-frequency disturbance waves. As an indirect support of this idea it should be noted that in the aforementioned model of Chang et al. (2002) coalescence of waves causes very similar transformation of the spectral structure (see Fig. 3 in that paper).

Within the same scenario of downstream evolution of PSD its shape does not qualitatively change with flow rates of the phases; increase of either gas or liquid flow rate leads mainly to stretching the PSD towards larger frequencies. Two important quantities could be extracted from the obtained spectral information. First, the frequency of the initial waves can be measured using the bottom string of each PSD matrix (shown by the thin solid blue lines in the bottom parts of Fig. 14). It is natural to suppose that these waves correspond to the waves of maximum growth rate appearing due to Kelvin–Helmholtz instability of film surface. Frequency of such waves is expected to monotonically grow with both gas and liquid flow rates. Nonetheless, at the largest gas velocities ($V_g \geq 43$ m/s) frequency of maximum power for $Re_L \geq 220$ suddenly falls and follows approximately the same dependence on gas velocity as $Re_L = 140$ does (see the solid symbols in Fig. 16).

Visual inspection of the spectra obtained at $x = 3$ mm showed that there exist local maxima in the range of high frequencies, such as those marked by (3) in the bottom parts of Fig. 14c and d. These maxima are located far from the main ‘plateau’ and their amplitude is much smaller than that of the main initial peak; so detection of them may be ambiguous. Assuming that these maxima correspond to the monotonically growing initial mode, we could make a very approximate estimation on how the frequency of this mode grows with V_g and Re_L . This estimation is shown by the dashed lines in Fig. 16 for $Re_L = 300$ and 400.

The reason of the non-monotonical behaviour of the initial frequency might be related to the interaction of the initial waves. With increasing gas velocity coalescence of neighbouring initial waves is more likely to occur earlier; this corresponds to the above-mentioned decrease in the distance over which the initial frequency peak is observed. Possibly, at the largest V_g and Re_L the first statistically

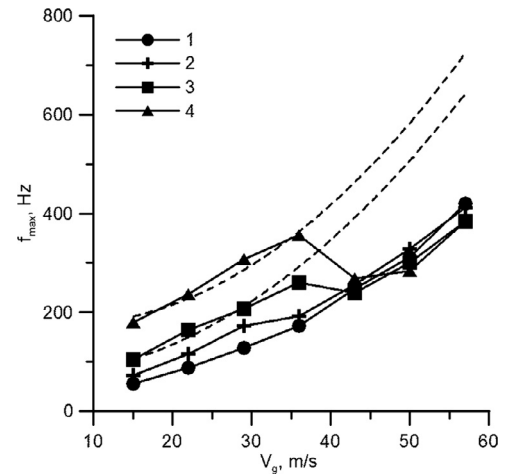


Fig. 16. Frequency of initial waves at $x = 3$ mm. Frequency of maximum power: $Re_L = 140$ (1); $Re_L = 220$ (2); $Re_L = 300$ (3); $Re_L = 400$ (4). Rough estimation of frequency of monotonically growing modes for $Re_L = 300$ and $Re_L = 400$ is shown by the dashed lines.

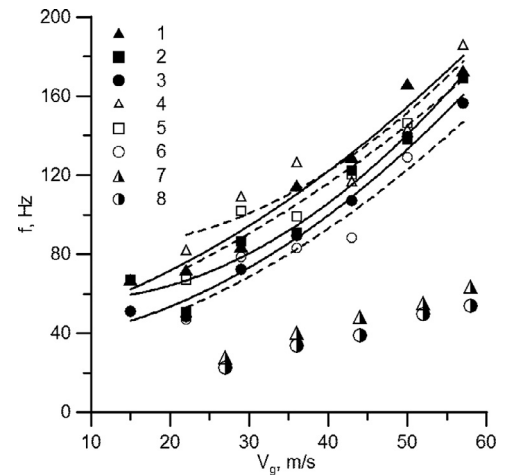


Fig. 17. Comparison of the disturbance waves frequency at $x = 40$ –100 mm (1–6) and $x = 500$ –620 mm (7–8), measured by spectral method (1–3) and by the automatic algorithm (4–8). $Re_L = 400$ (1, 4); 350 (7); 300 (2, 5); 220 (3, 6, 8). Parabolic approximations of data are shown by solid lines for spectral method and by dashed lines for the automatic counting.

significant rearrangement of initial waves occurs within the first few millimetres below the inlet. In particular, this might mean that the high-amplitude initial peaks marked by (1) in Fig. 14c and d actually appear due to the first stage of coalescence of higher-frequency initial waves.

The second quantity of interest is the initial frequency of the disturbance waves, which can be estimated as the frequency of maximum power averaged over the distance $x = 40$ –100 mm (i.e., the area where the stable growth of the average velocity was observed). This frequency was compared to the number of disturbance waves counted by the automatic algorithm described in section ‘‘Velocity of developing disturbance waves’’. The comparison results are given in Fig. 17. The frequencies obtained by the different methods are in relatively good agreement; this confirms that the frequency of maximum power in PSD can be used as an estimation of the frequency of the disturbance waves. In the same figure, the frequency measured with the automatic algorithm at $x = 500$ –620 mm is given. As it could be expected, the frequency drops significantly; about 60–70% of the disturbance waves appearing near the inlet will be absorbed during the first 0.5 m of the longitudinal distance.

Conclusions

Spatiotemporal evolution of liquid film in annular gas–liquid flow in the vicinity of the inlet was studied with laser-induced fluorescence technique. The main goal was to investigate the formation and the initial stages of development of the disturbance waves. Based on earlier observations, the disturbance waves were defined as the waves able to generate fast ripples (i.e. ripples appearing on the wave's rear slope and travelling over the wave's crest faster than the wave itself). It was shown that in the major part of the investigated range of flow conditions the disturbance waves appear and start to dominate in the wavy structure within the first 2–3 pipe diameters below the inlet. They are formed from the high-frequency initial waves, which appear at the inlet. Due to velocity variation, the initial waves undergo multiple coalescence acts, which coincide to the formation of the disturbance waves. Most likely, these high-frequency waves appear due to Kelvin–Helmholtz instability near the first point of contact of liquid film and high-velocity gas stream. Due to the effect of gas velocity the high-frequency waves grow much faster than the long-wave perturbations. Thus, coalescence of high-frequency waves is the quickest way in which the disturbance waves can be produced.

Similar phenomena were observed farther downstream. In particular, formation of ephemeral waves occurs due to coalescence of slow ripples behind the rear slopes of the disturbance waves. This process is more typical for high gas velocities. Another similar process was observed in the vicinity of transition to entrainment, at very high gas velocities and relatively small liquid flow rates. In this case, interaction of the primary waves (see Alekseenko et al., 2009) leads to formation of waves, covered by the fast ripples. Thus, the observed mechanism of formation of the disturbance waves is not limited by the specific case of the vicinity of a tangential slot inlet.

We suppose that the mechanism proposed is universal for any inlet configuration, if liquid is introduced as a film. If the liquid film is formed gradually (e.g., due to deposition of droplets from the gas-droplet flow, or due to condensation in steam flow), we expect the picture to be similar to what was observed at low film flow rates. First, the wavy regime with primary and secondary waves is going to form while the film is thin. With subsequent thickening of the film, the primary waves are expected to coalesce, producing disturbance waves.

It was found that near the inlet the average velocity of the disturbance waves linearly grows with the downstream distance if the disturbance waves dominate in the wavy structure. Estimated acceleration of disturbance waves in time linearly grows with gas velocity, starting from the velocity corresponding to the transition between 'dual-wave' and 'regular disturbance wave' regimes (see Webb and Hewitt, 1975). It was demonstrated that the observed increase in average velocity occurs mainly due to individual acceleration of disturbance waves. Energy transfer from high to low frequencies in process of downstream evolution of waves was observed using spectral analysis. This corresponds to visually observed process of coalescence of initial waves which leads to appearance of the disturbance waves.

Acknowledgements

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